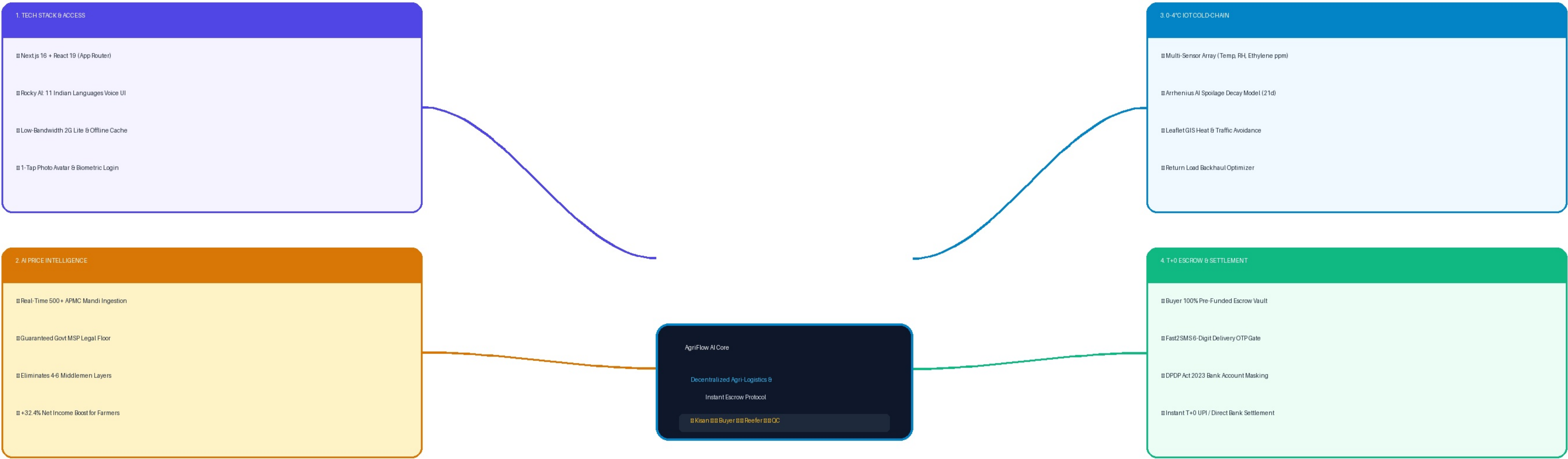
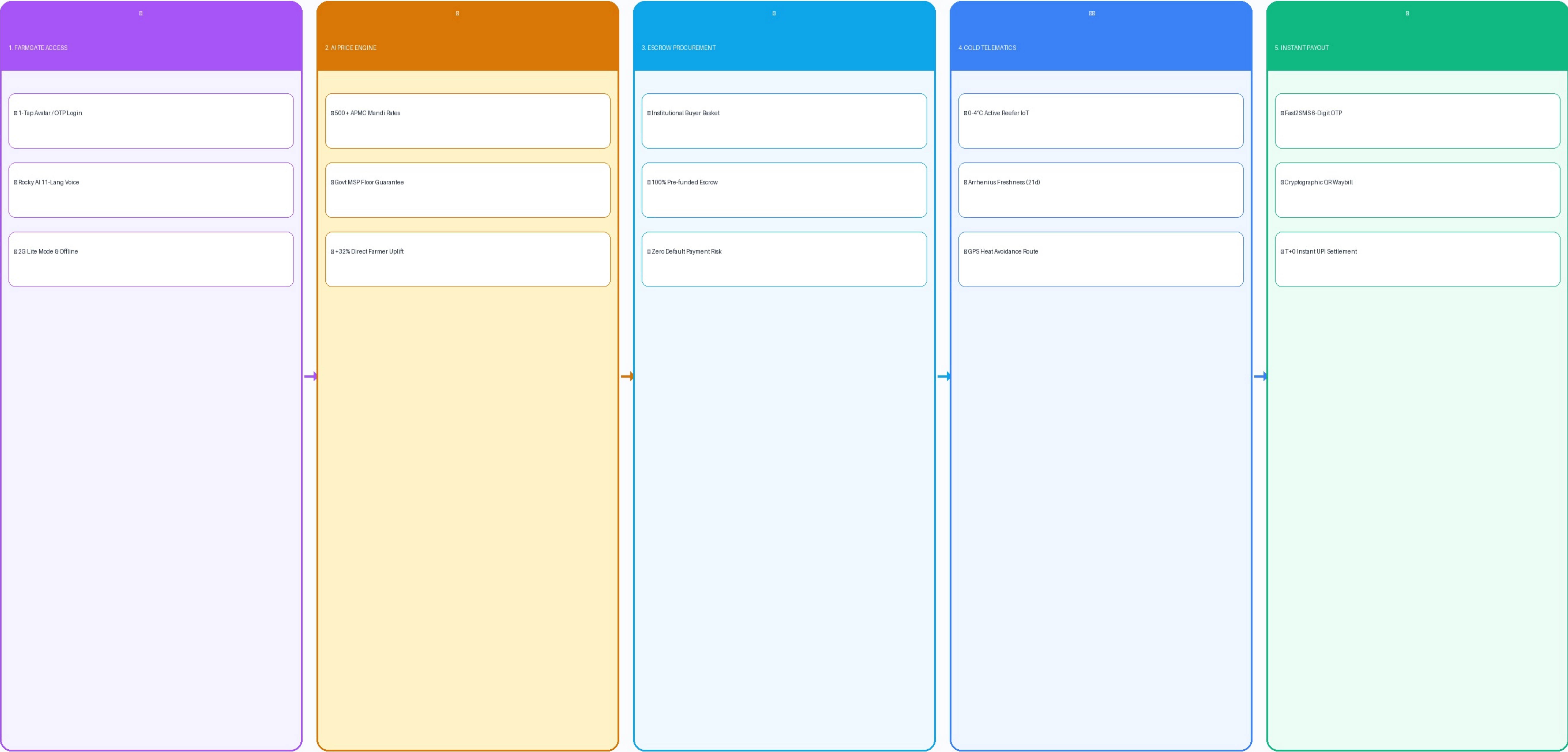
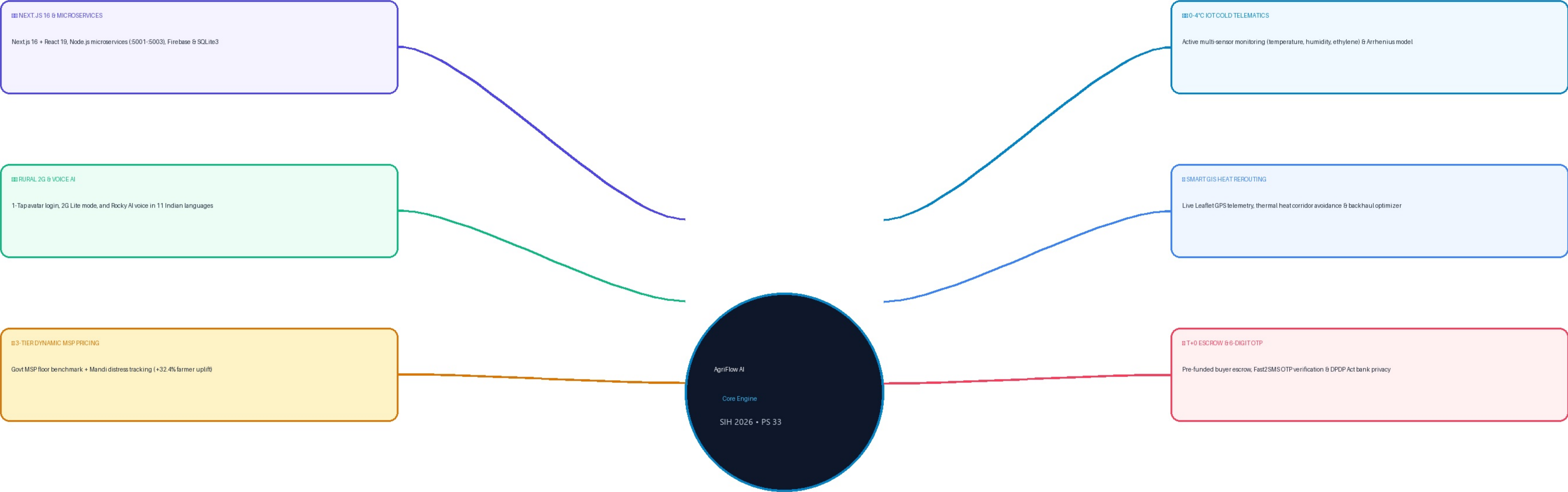




End-to-End AgriFlow AI Operation Flow • Quick Officer Review



Core Technical Modules at a Glance



Executive Summary for Fast Evaluation • Smart India Hackathon 2026

1. FULL-STACK INFRASTRUCTURE

▣ Next.js 16 (App Router) + React 19 Frontend

▣ 3 Decentralized Node.js Microservices (:5001 - :5003)

▣ Hybrid Storage: Firebase Firestore (NoSQL) + SQLite3

▣ Leaflet GIS Mapping & 528Hz SoundEngine

2. AI PRICING & FARMER UPLIFT

▣ Real-Time Scraping from 500+ APMC Mandis

▣ Legal Govt MSP Floor Benchmark Protection

▣ Direct Disintermediation (+32.4% Net Income Gain)

▣ Transparent Margin Split (80%+ to Farmer)

3. 0-4°C COLD-CHAIN TELEMATICS

▣ Multi-Sensor IoT: Temp (0-4°C), RH (85-90%), Ethylene

▣ Arrhenius Kinetic Model Tripling Shelf-Life (21 Days)

▣ Dynamic Thermal Corridor & Traffic Avoidance

▣ Return Load Backhaul Matching (Zero Empty Km)

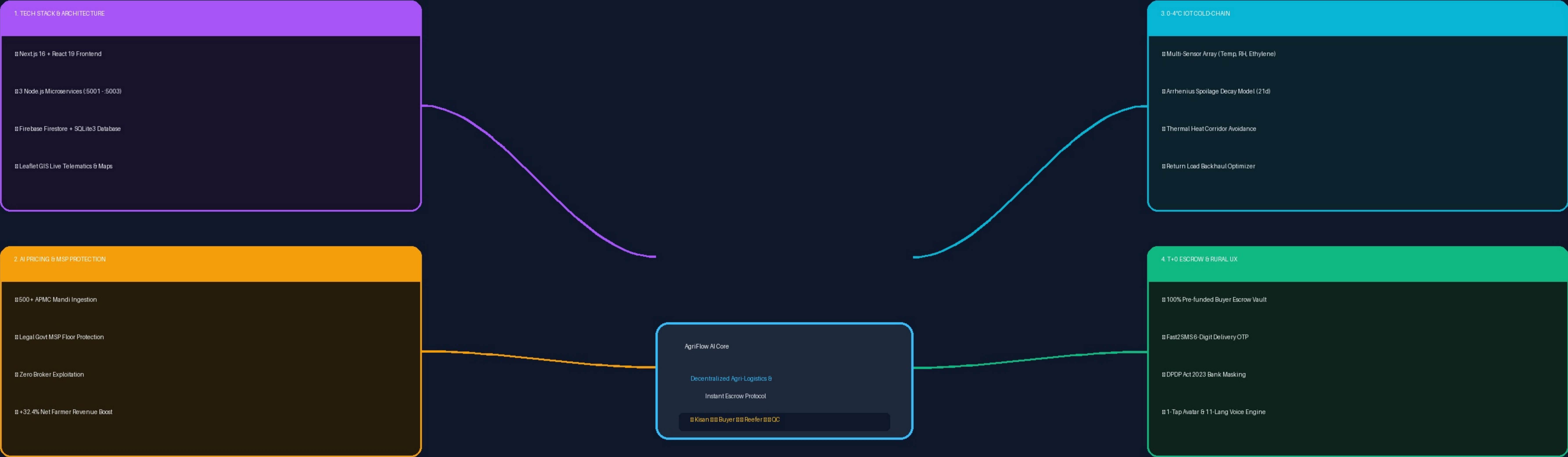
4. ESCROW TRUST & RURAL ACCESS

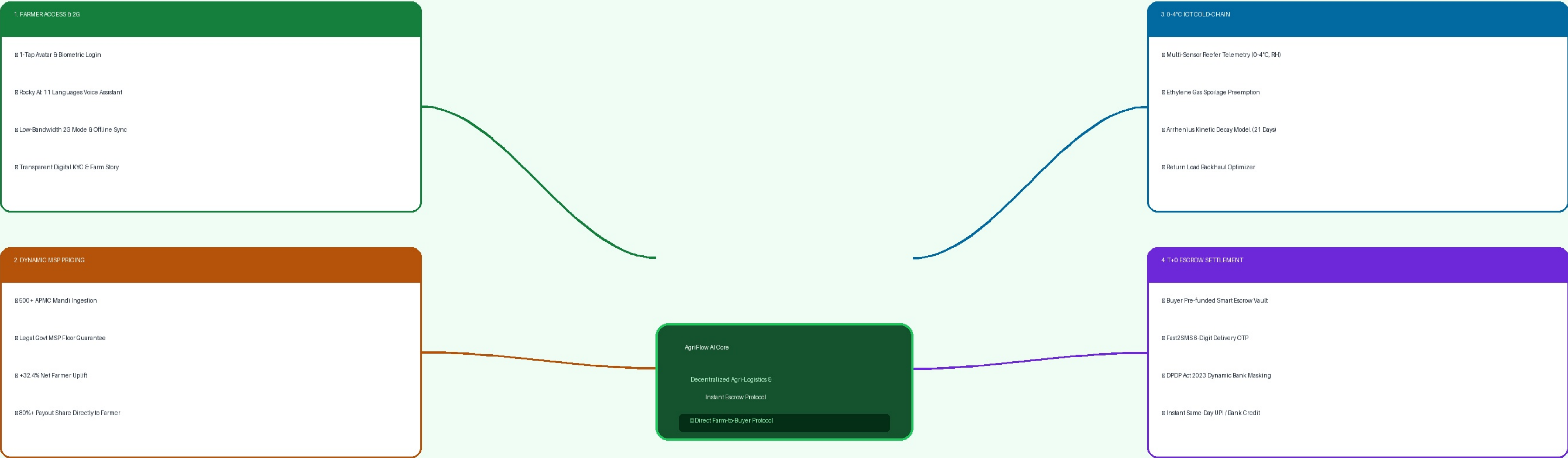
▣ Buyer 100% Pre-funded Escrow Vault (No Bad Debts)

▣ Fast2SMS6-Digit Delivery OTP Handover

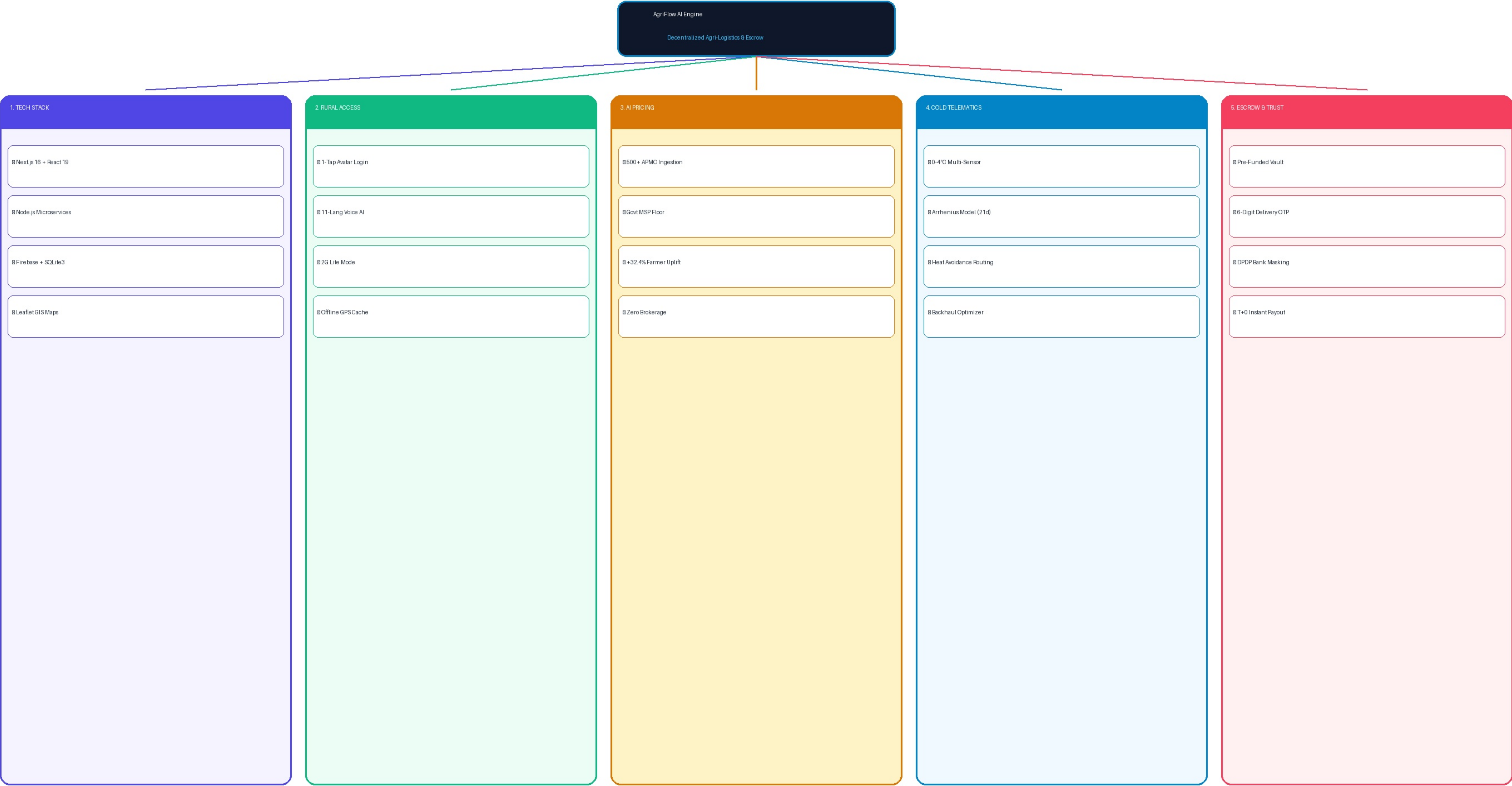
▣ DPDP Act 2023 Bank Masking (0000 0000 4412)

▣ 1-Tap Avatar Login & Rocky AI 11-Lang Voice Engine





Structured Architectural Breakdown for Fast Scanning



6 Core Value Drivers for Rapid Review

